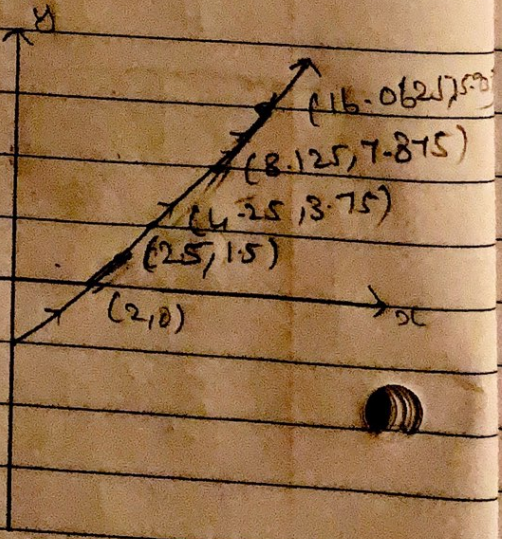
Quiz 18:-

C16:

$x = t + 1/t$ and $y = t - 1/t$

t	x	y
1	2	0
2	2.5	1.5
4	4.25	3.75
8	8.125	7.875
16	16.062	15.938



C17

$x = e^t \sin t$, $y = e^t \cos t$

$0 \le t \le 2\pi$

$$L = \int_{\theta_1}^{\theta_2} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$L = \int_{0}^{2\pi} \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\, dt$$

$$\frac{dx}{dt} = e^t \cos t + \sin t\, e^t$$

$$\frac{dy}{dt} = -e^t \sin t + \cos t\, e^t$$

$$L = \int_0^{2\pi} \sqrt{\left(e^t \cos t + \sin t\, e^t\right)^2 + \left(-e^t \sin t + \cos t\, e^t\right)^2}\, dt$$

$$\left(\frac{dx}{dt}\right)^2 = e^{2t}\cos^2 t + 2e^{2t}\sin t \cos t + \sin^2 t\, e^{2t}$$

$$\left(\frac{dy}{dt}\right)^2 = e^{2t}\sin^2 t - 2e^{2t}\sin t \cos t + \cos t\, e^{2t}$$

$$L = \int_0^{2\pi} \sqrt{e^{2t}\left(\cos^2 t + \sin^2 t + \sin^2 t + \cos^2 t\right)}\, dt$$

$$L = \int_0^{2\pi} \sqrt{2e^{2t}}\, dt$$

$$L = \sqrt{2}\int_0^{2\pi} e^t\, dt \Rightarrow L = \sqrt{2}\left[e^t\right]_0^{2\pi}$$

$$L = \sqrt{2}\left[e^{2\pi} - e^0\right]$$

$$\boxed{L = \sqrt{2}\left[e^{2\pi} - 1\right]}$$

C18

$$x^2 + y^2 = r^2$$

$$r = \frac{1}{2\cos\theta + 3\sin\theta}$$

$$r(1) = \left(\frac{1}{2\cos\theta + 3\sin\theta}\right)$$

$$= \frac{1}{4\cos^2\theta + 12\cos\theta\sin\theta + 9\sin^2\theta}$$

$$r = \frac{1}{2\cos\theta + 3\sin\theta}$$

$$\Rightarrow \quad 1 = \frac{1}{r(2\cos\theta + 3\sin\theta)}$$

$$\Rightarrow \quad 2r\cos\theta + 3r\sin\theta = 1$$

$$\Rightarrow \quad 2x + 3y = 1$$

$$\Rightarrow \quad \boxed{y = \frac{1 - 2x}{3}}$$

Q19:

$r = 2\cos(2\theta)$

$$A = \int_{\theta_1}^{\theta_2} \frac{1}{2} r^2 \, d\theta$$

$r = 2\cos(2\theta)$